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Chap 14 File

https://github.com/kkh3363/C_Lang

Contents

01 Create File

Files

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File Handling

- In C, you can create, open, read, and write to files by declaring a pointer of type FILE, and use the fopen() function:

```
FILE *fptr;
fptr = fopen(filename, mode);
```

- FILE is basically a data type, and we need to create a pointer variable to work with it (fptr).
- To open a file, use the fopen() function, which takes two parameters:

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```
FILE *fptr;
fptr = fopen(filename, mode);
```

Parameter	Description
filename	The name of the actual file you want to open (or create), like filename.txt
mode	A single character, which represents what you want to do with the file (read, write or append): w - Writes to a file a - Appends new data to a file r - Reads from a file

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- mode

Mode	Description
"w"	We use this mode to open or create a text file in writing mode. The data existing in the file is erased.
"r"	We use this mode for opening an existing file in reading mode. The file to be opened must exist.
"a"	We use this mode to open a text file in append mode. The existing data of the file is not erased as in “w” mode.
"w+"	We use this mode to open a text file in both reading and writing mode.
"a+"	We use this mode to open a text file in both reading and writing mode.
"r+"	We use this mode to open a text file in both reading and writing mode.
"wb"	We use this mode to open or create a binary file in writing mode.
"rb"	We use this mode to open a binary file in reading mode.
"ab"	We use this mode to open a binary file in append mode.
"wb+"	We use this mode to open a binary file in both reading and writing modes.
"rb+"	We use this mode to open a binary file in both reading and writing modes.
"ab+"	We use this mode to open a binary file in both reading and writing modes.

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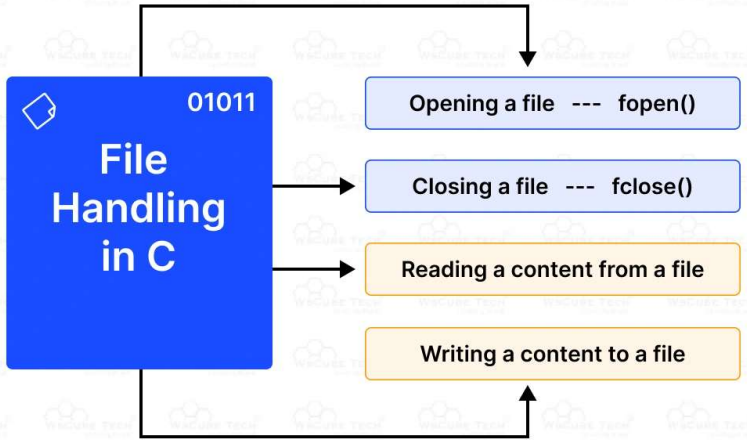
Create a File

- To create a file, you can use the *w* mode inside the fopen() function.
- The w mode is used to write to a file. However, if the file does not exist, it will create one for you:

```
FILE *fp;

// Create a file
fp = fopen("filename.txt", "w");

// Close the file
fclose(fp);
```



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Write To a File

- The w mode means that the file is opened for writing. To insert content to it, you can use the fprintf() function and add the pointer variable (fp in our example) and some text:

```
#include <stdio.h>

int main() {
  FILE *fp;

  // Open a file in writing mode
  fp = fopen(“test.txt", "w");

  // Write some text to the file
  fprintf(fp, "Some text");

  // Close the file
  fclose(fp);

  return 0;
}
```

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Append Content To a File

- If you want to add content to a file without deleting the old content, you can use the a mode.
- The a mode appends content at the end of the file:

```
#include <stdio.h>

int main() {
  FILE *fp;

  // Open a file in writing mode
  fp = fopen(“test.txt", “a");

  // Write some text to the file
  fprintf(fp, “\nHi everybody");

  // Close the file
  fclose(fp);

  return 0;
}
```

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- Read a File
  - To read from a file, you can use the **r** mode:
  - if file not exists, then fopen() return null.

```
fp = fopen("filename.txt", "r");
if ( fp == NULL )
   printf(“File not exists.”);
```

```
if ( (fp = fopen("filename.txt", "r")) == NULL )
   printf(“File not exists.”);
```

```
#include <stdio.h>

int main() {
  FILE *fp;

  // Open a file in read mode
  fp = fopen("filename.txt", "r");

  // Store the content of the file
  char myString[100];

  // Read the content and store it inside myString
  fgets(myString, 100, fp);

   printf("%s", myString); // Print file content

   fclose(fp); // Close the file

  return 0;
}
```

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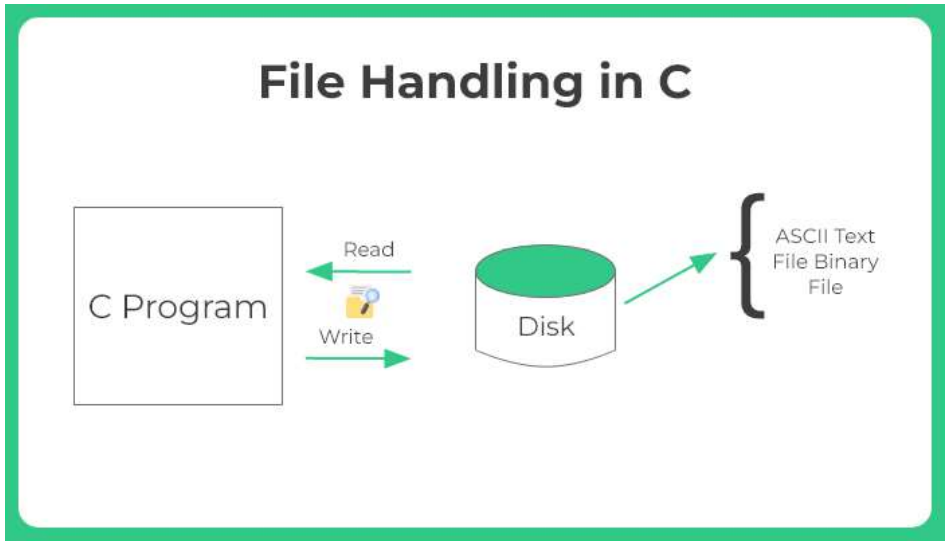
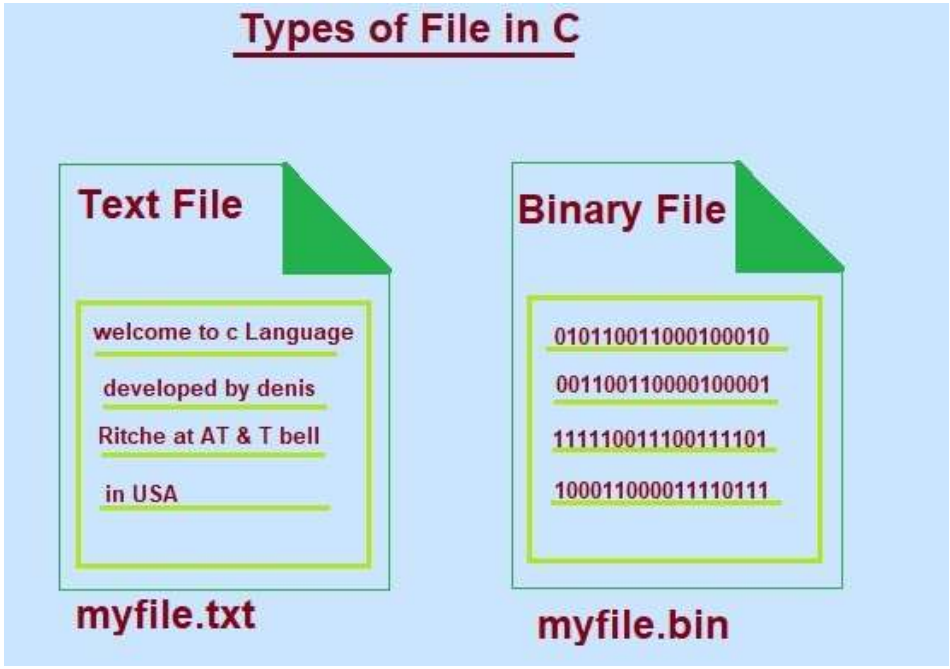
- Read a File





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- File Function

function	description.	mode
fprintf(fout, format, param)	write to file	Text
fscanf(fin, format, param)	read to file	Text
fputc(c, fout), putc(c, fout)	write to file (character)	Text
fgetc(fin), getc(fin)	read to file (character)	Text
fputs(s, fout)	write to file (string)	Text
fgets(s, n, fin)	read to file (string)	Text
fwrite(buf, size, n, fout)	write to file (binary block)	Binary
fread(buf, size, n, fin)	read to file (binary block)	Binary

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- Example Code
  - file copy

```
#define _CRT_SECURE_NO_WARNINGS
#include <stdio.h>

int main(int argc, char* argv[])
{
FILE* fpr ;
FILE* fpw;
char file;

if (argc != 3)
{
printf("usage %s : ", argv[0]);
printf("copyfile [source filename] [output filename]\n");
return 0;
}
```

Code ref

```
if ( (fpr = fopen(argv[1], "rb")) == NULL  )
return 0;
 fpw = fopen(argv[2], "wb");

while (!feof(fpr))
{
fread(&file, sizeof(char), 1, fpr);
fwrite(&file, sizeof(char), 1, fpw);
}

fclose(fpr);
fclose(fpw);

return 0;
}
```

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- Parsing Text or Data in C
  - String to Data field

Code ref

```
#include <stdio.h>
#include <string.h>

int main() {
    char data[] = "apple,banana,cherry";
    char *token = strtok(data, ",");

    while (token != NULL) {
        printf("Fruit: %s\n", token);
        token = strtok(NULL, ",");
    }
    return 0;
}
```

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- Mini project
  - phonebook
    - add function.
    - saveFile()
    - readFile()

    Code ref

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Thank You !